

AWS Solutions Architect Associate Study Guide

This study guide provides a comprehensive review of the AWS Solutions Architect Associate exam requirements, the AWS Well-Architected Framework, and specific architectural strategies for security, traceability, and identity management as outlined in the source materials.

Study Quiz: Short-Answer Questions

1. How does the AWS Well-Architected Framework define the "Sustainability" pillar and what is its primary focus?
2. Explain the difference between Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) in the context of the principle of least privilege.
3. What is the function of AWS Organizations and how does it assist in managing multiple AWS accounts?
4. What are the specific components and OUs created during the setup of an AWS Control Tower Landing Zone?
5. In the AWS single account policy evaluation logic, what is the significance of the "Default Deny" and "Explicit Deny"?
6. Describe the role of AWS Security Hub in the context of traceability and security event monitoring.
7. What is the difference between a Permission Boundary and a Session Policy?
8. How do cost-optimization tools differentiate between "Trending" and "Forecasting"?
9. What are the design principles associated with the "Operational Excellence" pillar of the Well-Architected Framework?

Quiz Answer Key

1. **Sustainability Pillar:** This pillar focuses on minimizing the environmental impacts of running cloud workloads, specifically targeting energy consumption and efficiency. Architects are encouraged to reduce resource usage by maximizing utilization and adopting more efficient hardware and software offerings.
2. **RBAC vs. ABAC:** RBAC grants access based on identity factors such as IAM users, group membership, or instance profiles. ABAC determines access based on properties or tags attached to users and resources, utilizing policy conditions to evaluate these attributes.
3. **AWS Organizations:** This service provides central management for multiple AWS accounts, offering account-level scoping and the creation of Organizational Units (OUs). It enables centralized billing and the implementation of central security policies across the entire organization.
4. **Control Tower Landing Zone:** A standard setup creates a ROOT structure with two OUs: Security and Sandbox. Within these, it establishes dedicated accounts for auditing (Audit) and log archiving (Log Archive) to ensure a secure foundation.

5. **Policy Evaluation Logic:** AWS evaluation begins with a "Default Deny," meaning no access is granted unless specified. An "Explicit Deny" in any applicable policy overrides any "Allow" instructions, immediately resulting in a final decision to deny access.
6. **AWS Security Hub:** Security Hub acts as a central repository that ingests findings from various services like Amazon GuardDuty, AWS Config, and Amazon Inspector. It provides a consolidated view of security alerts and compliance status across the environment.
7. **Boundaries and Session Policies:** A Permissions Boundary defines the maximum permissions a principal or account can have. A Session Policy is used to further limit permissions specifically while a user or service is assuming temporary credentials during a session.
8. **Trending vs. Forecasting:** Trending shows the variability in cost and usage over a specific required period of time to identify patterns. Forecasting provides the capability to estimate future costs based on historical data and current usage levels.
9. **Operational Excellence Principles:** This pillar emphasizes organizing teams around business outcomes, performing frequent and reversible changes, and refining procedures. It also stresses the importance of anticipating failure and learning from all operational events.

Essay Questions

1. **Synthesizing the Well-Architected Framework:** Discuss how the six pillars of the Well-Architected Framework (Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability) intersect to support the goal of translating business requirements into a resilient cloud architecture.
2. **The Hierarchy of Policy Evaluation:** Analyze the step-by-step logic AWS uses to evaluate policies (from SCPs to Session Policies). Explain why understanding this hierarchy is critical for a Solutions Architect when troubleshooting access issues in a multi-account environment.
3. **Centralized Governance via Control Tower:** Compare and contrast the benefits of using AWS Control Tower over manual AWS Organizations management. Focus on the roles of Guardrails, the Account Factory, and the Landing Zone in maintaining organizational compliance.
4. **Implementing Traceability and Remediation:** Describe a comprehensive security monitoring strategy using AWS services. Detail how logs move from ingestion (via CloudTrail or GuardDuty) to consolidation in Security Hub, and finally to automated remediation using EventBridge and Lambda.
5. **Identity Federation Strategies:** Evaluate the different use cases for SAML Federation, OpenID Connect (OIDC), and IAM Identity Center. Explain which federation method is most appropriate for a single-account setup versus a complex, multi-account enterprise environment.

Glossary of Key Terms

AWS Concepts, Security, and Architecture Terms

Term	Definition
ABAC	Attribute-Based Access Control; an authorization model that defines permissions based on attributes (tags).
AWS Config	A service that continuously monitors and evaluates configurations of AWS resources against desired standards.
AWS Control Tower	A service used to set up and govern a secure, multi-account AWS environment based on best practices.
CloudTrail	A service that tracks user activity and API usage for auditing and security purposes.
Default Deny	The base state of AWS permissions where all requests are denied unless an explicit "Allow" exists.
Explicit Deny	A policy statement that specifically forbids an action; it overrides any "Allow" statement in the evaluation logic.
GuardDuty	An intelligent threat detection service that monitors for malicious activity and unauthorized behavior.
Guardrails	High-level rules in AWS Control Tower that provide ongoing governance for an entire AWS environment.
IAM Identity Center	A service that helps centralize identity management for multiple AWS accounts and applications.

Least Privilege	The security principle of granting only the minimum permissions necessary to perform a specific task.
Mechanical Sympathy	A design principle of Performance Efficiency involving understanding how cloud resources work to use them effectively.
Operational Excellence	The pillar focused on running and monitoring systems to deliver business value and continually improving processes.
RBAC	Role-Based Access Control; an authorization model based on the identity of the user or group.
SCP	Service Control Policy; a type of organization policy used to manage permissions in an organization.
Security Hub	A cloud security posture management service that performs security checks and aggregates alerts.
Sustainability	The pillar focused on environmental impacts, resource efficiency, and reducing the downstream impact of cloud workloads.